NASA SPACE APPS CHALLENGE 2026 · ALMATY, KAZAKHSTAN

# Field Shift KZ

Crop rotation decision support for Kazakhstan farms, built on NASA Earth observations

CHALLENGE

**Field Shift: Adapting Farms with NASA Data**

TEAM DDT

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Live demo: stalker-a-field-shift-kz-app-xokyoc.streamlit.app · Code: github.com/Stalker-a/field-shift-kz

THE PROBLEM

# Over half of Kazakhstan's cropland is one crop

**57%**

of all sown area in 2024 was wheat, and 95% of that wheat was spring wheat

**0.87 t/ha**

average grain yield in the 2012 drought, vs 1.85 t/ha in 2011; the wheat harvest fell 57%

**33 → 42**

hot days (30 °C or more) per season in Aktobe: 1981–2000 → 2006–2025

**The climate is shifting unevenly.** Kostanay's growing season barely warmed but got 13% wetter, while the west gained almost a third more hot days. One “wheat ×3 + fallow” recipe no longer fits every field.

Sources: Bureau of National Statistics of Kazakhstan, refined 2024 sown areas (tables 1, 6); Ministry of Agriculture of Kazakhstan and USDA FAS, 2012; NASA POWER, Field Shift KZ analysis

THE SOLUTION

# Field Shift KZ: from NASA data to a field's rotation

1. **NASA POWER data**

   daily records 1981–2025 for the field's location

2. **8 field indicators**

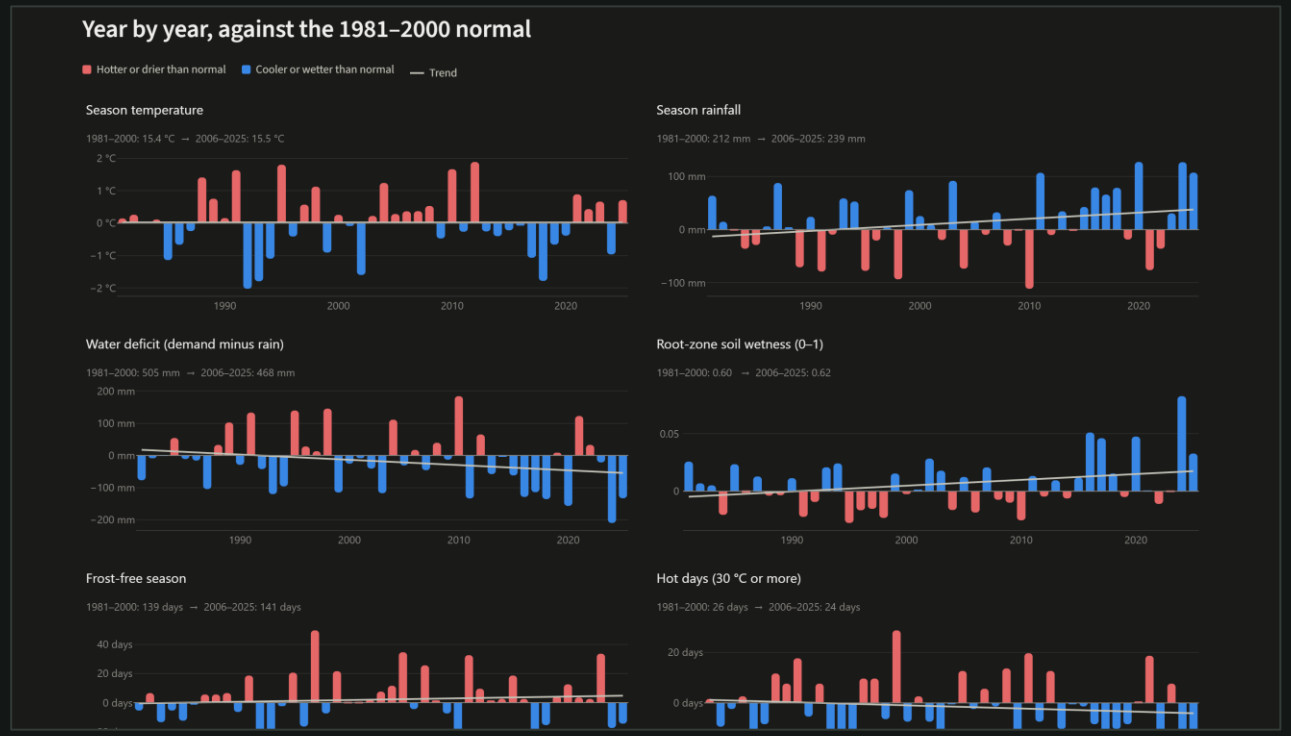
   frost-free season, hot days, water deficit and more

3. **Fit of 16 crops**

   today and on the 2035 trend

4. **3 four-year rotations**

   soil, water, income, resilience; the farmer sets the weights




Climate shift tab: Kostanay, 2006–2025 vs 1981–2000

**Live demo:** stalker-a-field-shift-kz-app-xokyoc.streamlit.app

DEMO · KOSTANAY, CHERNOZEM, RAINFED

# Legumes instead of fallow: 53% → 83%

**Today**

Wheat → wheat → wheat → fallow

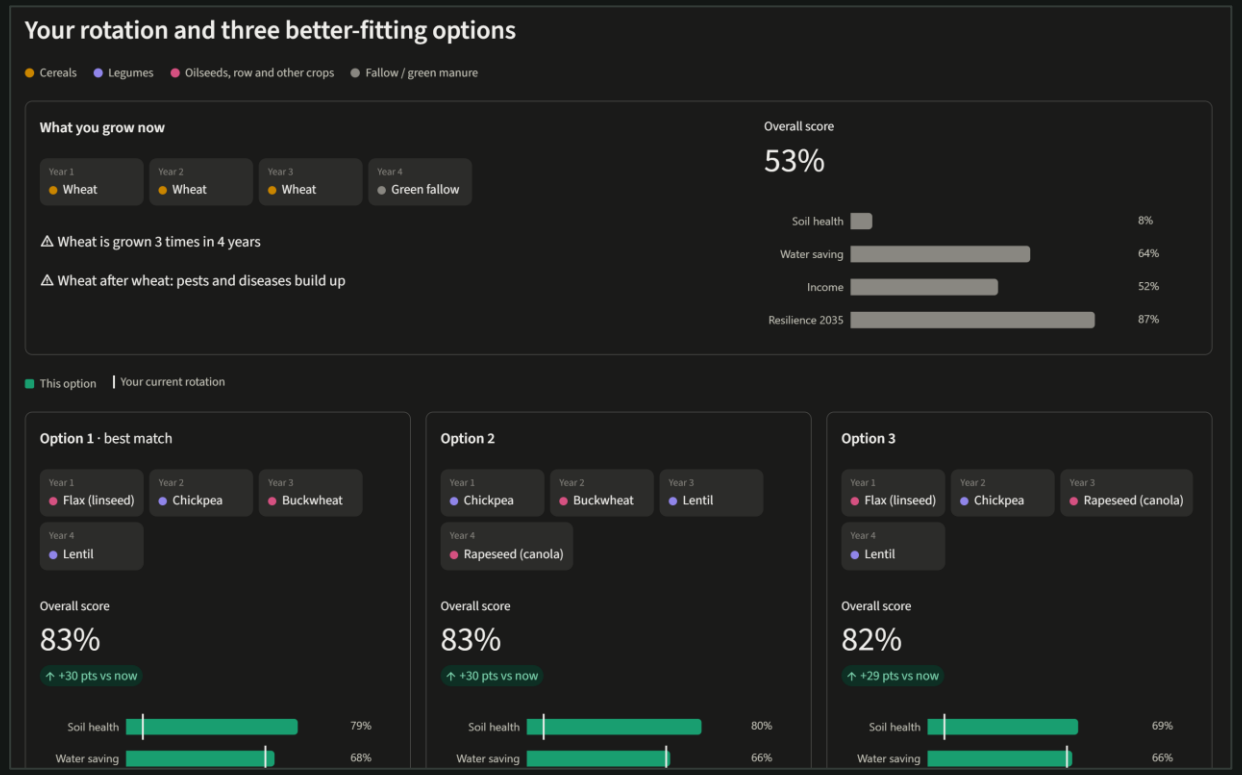
**53%**

**Option 1**

Flax → chickpea → buckwheat → lentil

**83%**

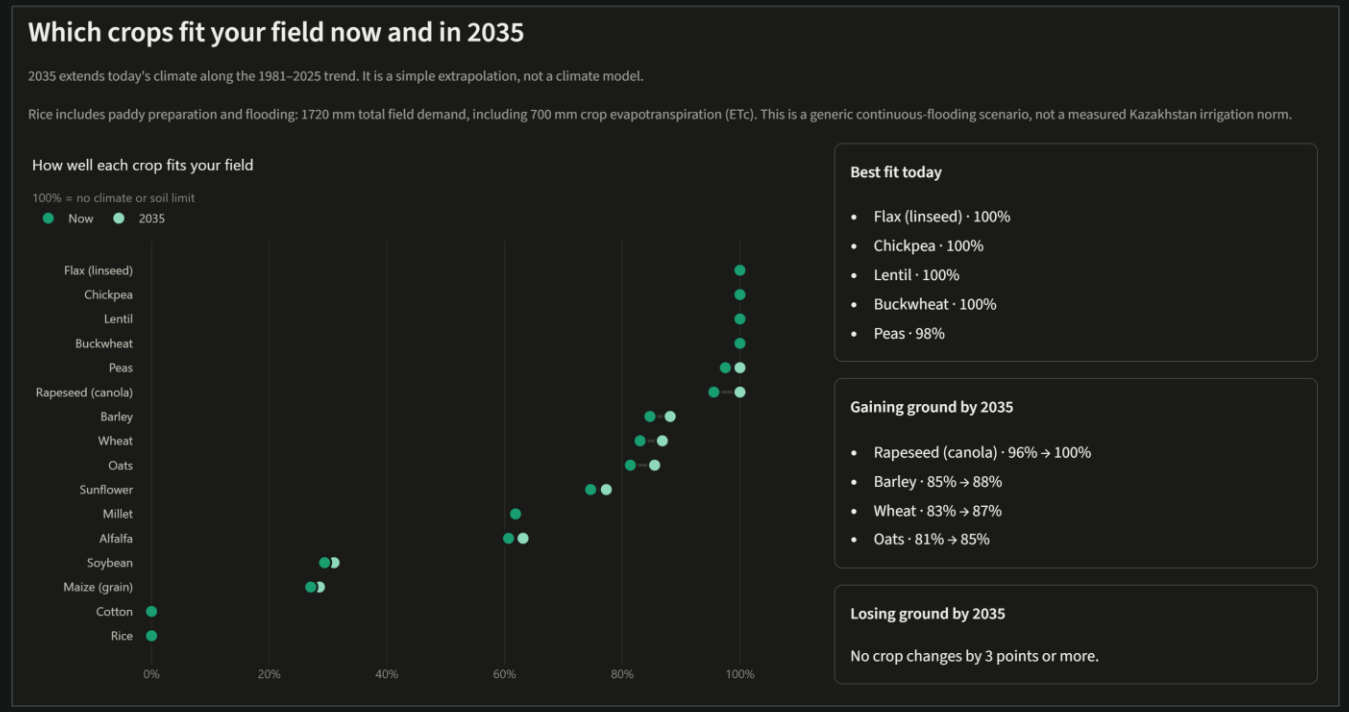
| | | |
|---|---|---|
| Soil health | 8% | **79%** |
| Water saving | 64% | **68%** |
| Income | 52% | **86%** |
| Resilience 2035 | 87% | **100%** |




Wheat after lentil yields ~88% of wheat after fallow (Barayev Center, Akmola region, 2018–2020), and Kazakhstan sowed 25% more pulses in 2024 than in 2023 (Bureau of National Statistics).

DATA AND LIMITS

# NASA data and honest limits




Crop fit tab: crop suitability now and in 2035, Kostanay

## Data

NASA POWER daily API, 1981–2025, no key needed

T2M, T2M_MAX, T2M_MIN, PRECTOTCORR, GWETROOT

Reference evapotranspiration: Hargreaves (FAO-56)

Crop parameters checked against FAO and the Barayev Center (SOURCES.md)

## Limits

2035 extends the observed trend; it is not a climate model

Compares options; it does not forecast yields

The farmer picks the soil type for now

“Income” is a relative index, not market prices

NEXT STEPS

# What comes next

### 1 Telegram bot

in Kazakh and Russian: share a field location, get three rotations

### 2 Satellite NDVI

MODIS, Landsat, Sentinel-2: what actually grew on the field

### 3 Soil from SoilGrids

humus and water capacity by coordinates, no manual input

### 4 Prices and calibration

market prices instead of an income index; validation with Kazakh research institutes

**Try it:** stalker-a-field-shift-kz-app-xokyoc.streamlit.app **Thank you!**